Year 10 Circle Geometry - Homework Answers

Revision | Warm-up

1 Year 8 Exercise 6C Q6

a The base angles of an isosceles triangle are equal, so $a + a + 50 = 180$ (angle sum of a triangle). Then $2a = 130$, so $a = 65$.

b The equal sides meet at the $100°$ angle, so $a°$ is a base angle and the other base angle is also $a°$. Then $a + a + 100 = 180$ and $2a = 80$, so $a = 40$.

c The equal sides meet at the $28°$ angle, so $a + a + 28 = 180$ and $2a = 152$, so $a = 76$.

2 Year 8 Exercise 6C Q10

a $\angle ACB = 60°$ (angle in an equilateral triangle)
$\angle DCE = \angle ACB = 60°$ (vertically opposite angles)
$\therefore a = 60$

b $\angle ACB + 22° + 29° = 180°$ (angle sum of $\triangle ABC$)
$\therefore \angle ACB = 129°$
$a = 360 - 129 = 231$ (angles in a revolution)

e $\angle CAB = \angle CBA = a°$ (equal angles are opposite equal sides)
$\angle ACB = \angle DCE = 49°$ (vertically opposite angles)
$a + a + 49 = 180$ (angle sum of $\triangle ABC$)
$2a = 131$
$a = 65.5$

3 Year 8 Exercise 8E Q3

a SAS (two sides, 3 cm and 4 cm, and the included angle $110°$) **b** SSS (three pairs of equal sides)

c RHS (right angle, hypotenuse 7 cm, second side 4 cm) **d** AAS (angles $50°$ and $80°$, and the 3 cm side opposite the $50°$ angle in both)

4 Year 8 Exercise 8G Q3

a

$$\frac{DE}{AB} = \frac{10}{5} = 2$$
$$\frac{DF}{AC} = \frac{24}{12} = 2$$
$$\frac{EF}{BC} = \frac{26}{13} = 2$$

$\therefore \triangle ABC \,|||\, \triangle DEF$ (three sides in proportion)

b

$\angle A = \angle D$ and $\angle B = \angle E$

$\therefore \triangle ABC \,|||\, \triangle DEF$ (two pairs of equal angles)

c

$$\frac{DE}{AB} = \frac{10}{5} = 2$$
$$\angle D = \angle A$$
$$\frac{DF}{AC} = \frac{6}{3} = 2$$

$\therefore \triangle ABC \,|||\, \triangle DEF$ (two sides in proportion and the included angles equal)

d

$$\angle D = \angle A = 90°$$
$$\frac{EF}{BC} = \frac{26}{13} = 2$$
$$\frac{DF}{AC} = \frac{22}{11} = 2$$

$\therefore \triangle ABC \,|||\, \triangle DEF$ (hypotenuse and a second side in proportion)

5 Year 8 Exercise 8H Q5

In $\triangle ABC$ and $\triangle BCD$:
$\angle ABC = \angle BCD$ (angles in a square are equal)
$AB = BC$ (sides of a square are equal)
BC is common.
$\therefore \triangle ABC \equiv \triangle BCD$ (SAS)
$\therefore AC = BD$ (matching sides in congruent triangles are equal)
and the diagonals of a square are equal in length.

Year 10 Circle Geometry - Homework Answers

Lesson C1 | Chords, radii and angles

Lesson Examples

C1 **Tutor demo.**

1 **a** Tangent **b** Chord (a diameter) **c** Secant **d** None (its ends are not on the circle) **e** Chord **f** None (a radius)

2 Diameter $= 2 \times 6 = 12$ cm. Circumference $= 2\pi r = 2 \times \pi \times 6 = 12\pi$ cm. Area $= \pi r^2 = \pi \times 6^2 = 36\pi$ cm^2.

Your turn.

1 **a** Chord **b** None (it misses the circle) **c** Tangent **d** Secant **e** None (a radius) **f** Chord (a diameter)

2 Diameter $= 2 \times 4.5 = 9$ cm. Circumference $= 2 \times \pi \times 4.5 = 9\pi$ cm. Area $= \pi \times 4.5^2 = 20.25\pi$ cm^2, that is, $\frac{81}{4}\pi$ cm^2.

C2 **Tutor demo.**

a The radius is perpendicular to the chord, so it bisects the chord. Therefore $u = 3.5$ cm.

b The radius bisects the chord, so it is also perpendicular to the chord. Therefore $q + 35° = 90°$, so $q = 90° - 35° = 55°$.

c Let the chord be AB, with centre O and midpoint M. The interval OM joining the centre to the midpoint is perpendicular to AB, and $AM = \frac{1}{2} \times 24 = 12$ cm. In right-angled $\triangle OAM$, by Pythagoras' theorem, $OM^2 = OA^2 - AM^2 = 13^2 - 12^2 = 25$, so the distance is $OM = 5$ cm.

Your turn.

a The radius is perpendicular to the chord, so it bisects the chord. Therefore $v = 4.5$ m.

b The radius bisects the chord, so it is also perpendicular to the chord. Therefore $p + 22° = 90°$, so $p = 90° - 22° = 68°$.

c With M the midpoint of the chord AB, $OM \perp AB$ and $AM = \frac{1}{2} \times 16 = 8$ cm. By Pythagoras' theorem, $OM^2 = 10^2 - 8^2 = 36$, so the distance is $OM = 6$ cm.

C3 **Tutor demo.**

a The chords are equal (both 9 m), so the angles they subtend at the centre are also equal. Therefore $y = 48°$.

b The angles subtended at the centre are equal (both $105°$), so the chords which subtend them are equal. Therefore $x = 3$ cm.

c Two sides of the triangle are radii, so it is isosceles and its base angles are equal (base angles of an isosceles triangle). Each is $\frac{180° - 48°}{2} = 66°$ (angle sum of a triangle).

Your turn.

a The chords are equal (both 5 m), so the angles they subtend at the centre are equal. Therefore $z = 62°$.

b The angles subtended at the centre are equal (both $130°$), so the chords which subtend them are equal. Therefore $x = 2.5$ cm.

c The triangle has two radii as sides, so its base angles are equal. Each is $\frac{180° - 62°}{2} = 59°$.

C4 **Tutor demo.**

a $\angle APO = 28°$ (base angles of isosceles $\triangle AOP$, since $OA = OP$ are radii)
$\angle APB = 90°$ (angle in a semicircle)
$\angle BPO = 90° - 28° = 62°$ ($\angle APO + \angle BPO = \angle APB$)
$\angle B = 62°$ (base angles of isosceles $\triangle BOP$, since $OB = OP$ are radii)

b By Pythagoras' theorem, $AB^2 = 6^2 + 8^2 = 100$, so $AB = 10$ cm. The circle with diameter AB passes through C (converse of the angle in a semicircle theorem), so its centre M is the same distance from A, B and C: $CM = MA = \frac{1}{2} \times 10 = 5$ cm.

Your turn.

a $\angle BPO = 57°$ (base angles of isosceles $\triangle BOP$, since $OB = OP$ are radii)
$\angle APB = 90°$ (angle in a semicircle)
$\angle APO = 90° - 57° = 33°$ ($\angle APO + \angle BPO = \angle APB$)
$\angle A = 33°$ (base angles of isosceles $\triangle AOP$, since $OA = OP$ are radii)

b By Pythagoras' theorem, $AB^2 = 5^2 + 12^2 = 169$, so $AB = 13$ cm. M is the centre of the circle through A, B and C, so $CM = \frac{1}{2} \times 13 = 6.5$ cm.

Homework

EASY

1 Maths Pathway, Chords, Secants and Tangents, Q1(g)–(l)

g Chord **h** Secant **i** Tangent **j** None **k** Tangent **l** Secant

2 Maths Pathway, Chords, Secants and Tangents, Q3(a)

The angles are equal, so the chords they are subtended by are also equal. Therefore $x = 2$ cm.

3 Maths Pathway, Chords, Secants and Tangents, Q5(a)

The radius is perpendicular to the chord, so it bisects it. Therefore $u = 2$ m.

4 Maths Pathway, Arcs and Angles in Circles, Q7

a The angle at x is subtended at a point on the circle by a diameter, so $x = 90°$ (angle in a semicircle).

b The diameter is a straight line through the centre, so $y = 180°$ (straight angle), and $z = 90°$ (angle in a semicircle).

c The angle at x is subtended by a diameter, so $x = 90°$ (angle in a semicircle).

The sheet's answers also write these as $2x = 180°$, using the angle at the centre, which is Lesson C2.

5 Maths Pathway, Arcs and Angles in Circles, Q8(a)

$a = 90°$, because the angle marked a is subtended by the diameter at a point on the circle.

MEDIUM

6 Maths Pathway, Chords, Secants and Tangents, Q3(d)

The chords of the larger circle are equal (both 6 m), so the angles they subtend at the centre are also equal. Therefore $t = 65°$.

The angles are equal, so the chords which the angles are subtended by in the smaller circle are also equal. Therefore $w = 3$ m.

7 Maths Pathway, Chords, Secants and Tangents, Q6(a)

A and C lie on the circle with centre O, so AC is a chord of that circle. Since the radius OR is perpendicular to AC, it bisects AC: $BC = AB = 4$ mm.

B and D lie on the circle with centre P, so BD is a chord of that circle. Since the radius PS is perpendicular to BD, it bisects BD: $CD = BC = 4$ mm.

$AD = AB + BC + CD = 4 + 4 + 4 = 12$ mm

8 Maths Pathway, Chords, Secants and Tangents, Q6(b)

Since $AH = HD$, GH joins the centre to the midpoint of the chord AD, so $\angle GHD = 90°$.

$\angle GHI + \angle IHD = \angle GHD$, so $\angle GHI + 52° = 90°$ and $\angle GHI = 38°$.

Since $BI = IF$, GI joins the centre to the midpoint of the chord BF, so $\angle GIH = 90°$.

$\angle GHI + \angle IGH + \angle GIH = 180°$ (angle sum of $\triangle GHI$), so $38° + \angle IGH + 90° = 180°$ and $\angle IGH = 52°$.

9 AMSI TIMES Circle Geometry, Exercise 2

The source's answer: use simple trigonometry in $\triangle AMO$. In full: M is the midpoint of the chord AB, so $OM \perp AB$ and $\triangle AMO$ is right-angled at M, with hypotenuse $OA = 1$.

a $\sin\theta = \dfrac{\text{opposite}}{\text{hypotenuse}} = \dfrac{AM}{OA} = \dfrac{AM}{1}$, so $AM = \sin\theta$.

b $\cos\theta = \dfrac{\text{adjacent}}{\text{hypotenuse}} = \dfrac{OM}{OA} = \dfrac{OM}{1}$, so $OM = \cos\theta$.

10 AMSI TIMES Circle Geometry, Exercise 1

a Every translation of any distance along ℓ 'maps' ℓ onto itself. Let O be any point on ℓ. Then reflection in the line m through O perpendicular to ℓ, and rotation of $180°$ about O, both map ℓ onto itself (and are identical in their action on the points of ℓ). Reflection of the plane in the line ℓ fixes the line ℓ, because it fixes every point on ℓ.

b The translation of the plane moving P to Q maps ℓ onto itself, and maps P to Q. Reflection in the perpendicular bisector of the interval PQ maps ℓ onto itself, and exchanges the points P and Q. (Rotation of $180°$ about the midpoint of PQ does the same.)

11 Maths Pathway, Chords, Secants and Tangents, Q2

a In ΔAOB and ΔCOD, we compare corresponding angles and sides.

$AO = CO$ (Radii of the same circle)

$OB = OD$ (Radii of the same circle)

$\angle AOB = \angle COD$ (Given)

Therefore $\Delta AOB \equiv \Delta COD$ by SAS.

b Since AB and CD are corresponding sides in congruent triangles, then $AB = CD$.

12 AMSI TIMES Circle Geometry, Exercise 4

At all times, the front of the building is the hypotenuse of a right-angled triangle whose third vertex is the photographer. Hence the circle with diameter, the front of the building, always passes through the photographer, and his possible positions are the points on the semicircle in front of the building.

HARD

13 Maths Pathway, Chords, Secants and Tangents, Q6(c)

AB is a diameter of the smaller circle and C is on that circle, so $\angle ACB = 90°$ (angle in a semicircle).

By Pythagoras' theorem in $\triangle ACB$: $AC^2 + BC^2 = AB^2$, so $AC^2 + 4^2 = 5^2$, $AC^2 = 9$ and $AC = 3$ m.

A and D are on the larger circle, so AD is a chord of it. BE is a diameter, so it passes through the centre, and $BC \perp AD$. The line from the centre perpendicular to a chord bisects it, so $CD = AC = 3$ m.

14 AMSI TIMES Circle Geometry, Exercise 3

All radii of a circle are equal, so $OA = OB = OP = OQ$, so the quadrilateral $APBQ$ is a rectangle because its diagonals are equal and bisect each other. Hence $\angle APB$ is a right angle.

In detail: the diagonals AB and PQ are both diameters, so each has length twice the radius, and O is the midpoint of each.

15 AMSI TIMES Circle Geometry, Exercise 5

At all times, the triangle formed by the plank, the wall and the floor is right-angled with hypotenuse of length ℓ metres. By the converse theorem above, the midpoint of the plank is therefore always $\dfrac{\ell}{2}$ metres from the corner. Hence the midpoint traces out a quadrant of the circle with centre at the corner and radius $\dfrac{\ell}{2}$ metres.

Year 10 Circle Geometry - Homework Answers

Lesson C1 | Chords, radii and angles

Past-Paper Homework - Worked Solutions

PP 1 **Sydney Boys High School, 2020, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics Stage 5.3, Q40(b) I**

In the cross section, X and Y lie on the circle with centre A, so XY is a chord of that circle. AB bisects XY, so the line AB passes through the centre A and the midpoint of the chord XY.

The line from the centre of a circle to the midpoint of a chord is perpendicular to the chord. $\therefore AB \perp XY$.

The marking scheme gives the mark for using this circle theorem. A congruent-triangle proof ($AX = AY$, $MX = MY$, AM common, SSS) is also correct but long for 1 mark.

PP 2 **Sydney Boys High School, 2022, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part A Q9**

The pentagon is regular, so its five sides are equal chords of the circle. Equal chords subtend equal angles at the centre, and these five angles make a revolution, so

$$\angle AOB = \frac{360^\circ}{5} = 72^\circ.$$

AOF is a diameter, a straight line, so $\angle BOF = 180^\circ - 72^\circ = 108^\circ$.

$OB = OF$ (radii), so $\triangle OBF$ is isosceles and $\angle OBF = \dfrac{180^\circ - 108^\circ}{2} = 36^\circ$.

The school's solution reaches the same answer through the interior angle of the pentagon: $\angle EAB = 108^\circ$, reflex $\angle EOB = 216^\circ$, so $\angle FOB = 108^\circ$ and $\angle OBF = 36^\circ$.

PP 3 **Sydney Boys High School, 2021, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part E (a)**

O lies on AD, so $\angle ODC = \angle ADC = 35^\circ$. In $\triangle OCD$, $OC = OD = 9$ cm (radii), so $\angle OCD = 35^\circ$ (base angles of an isosceles triangle).

$\angle AOC = 35^\circ + 35^\circ = 70^\circ$ (exterior angle of $\triangle OCD$).

The shaded segment is the sector OAC minus $\triangle AOC$.

Area of sector $= \dfrac{70}{360} \times \pi \times 9^2 = 49.480\ldots$ cm^2

Area of $\triangle AOC = \frac{1}{2} \times 9 \times 9 \times \sin 70^\circ = 38.057\ldots$ cm^2

Area of segment $= 49.480\ldots - 38.057\ldots = 11.4$ cm^2 (to 3 significant figures)

The school's solution finds $\angle AOC = 70^\circ$ with the angle at the centre theorem of Lesson C2. The isosceles triangle and the exterior angle give it with this lesson's tools.

PP 4 **Merewether High School, 2023, Year 10 Assessment Task 4, Mathematics, Section III Part B Q2**

In $\triangle OJI$ and $\triangle OJK$:
$OI = OK$ (equal radii)
OJ is common
$\angle IJO = \angle KJO = 90°$ ($OJ \perp IK$)
$\therefore \triangle OJI \equiv \triangle OJK$ (RHS)
$\therefore \angle IOJ = \angle KOJ$ (matching angles of congruent triangles)
$\therefore OJ$ bisects $\angle IOK$.

Year 10 Circle Geometry - Homework Answers

Lesson C2 | Angles at the centre and circumference

Lesson Examples

C1 **Tutor demo.**

a The $35°$ angle at the circumference and x at the centre stand on the same arc, so $x = 2 \times 35° = 70°$.

b The $116°$ angle at the centre and y at the circumference stand on the same arc, so $2y = 116°$ and $y = 58°$.

c The $150°$ angle stands on the major arc, and so does the reflex angle x at the centre, so $x = 2 \times 150° = 300°$.

Your turn.

a $x = 2 \times 32° = 64°$ (angle at the centre is twice the angle at the circumference on the same arc)

b $2y = 96°$, so $y = 48°$.

c The reflex angle at the centre stands on the same major arc as the $145°$ angle, so $x = 2 \times 145° = 290°$.

C2 **Tutor demo.**

a x and the $62°$ angle are at the circumference and stand on the same arc, so $x = 62°$.

b y and the $41°$ angle stand on the same arc (between the top point and the right end of the diameter), so $y = 41°$.

Your turn.

a $x = 53°$ (angles at the circumference standing on the same arc) **b** $y = 28°$ (angles on the same arc)

C3 **Tutor demo.**

a Opposite angles of a cyclic quadrilateral are supplementary, so $x = 180° - 64° = 116°$.

b a and the $128°$ angle are opposite angles of a cyclic quadrilateral, so $a = 180° - 128° = 52°$.

c The exterior angle of a cyclic quadrilateral equals the opposite interior angle, so $x = 71°$.

Your turn.

a $x = 180° - 48° = 132°$ **b** $a = 180° - 135° = 45°$ **c** $x = 83°$ (exterior angle of a cyclic quadrilateral)

C4 **Tutor demo.**

$\angle D = 360° - 97° - 78° - 83° = 102°$ (angle sum of a quadrilateral). Then $\angle A + \angle C = 97° + 83° = 180°$, so the opposite angles are supplementary and $ABCD$ is cyclic (test for a cyclic quadrilateral).

Your turn.

$\angle S = 360° - 88° - 96° - 93° = 83°$. Then $\angle P + \angle R = 88° + 93° = 181°$. If $PQRS$ were cyclic, its opposite angles would add to $180°$, so $PQRS$ is not cyclic.

C5 **Tutor demo.**

The diameter is $2R = \dfrac{a}{\sin A} = \dfrac{12}{\sin 30°} = \dfrac{12}{\frac{1}{2}} = 24$ cm, so the radius is $R = 12$ cm.

Your turn.

$2R = \dfrac{9}{\sin 50°}$, so $R = \dfrac{9}{2\sin 50°} = 5.87$ cm (to two decimal places).

Homework

EASY

1 Maths Pathway, Arcs and Angles in Circles, Q1

a Yes. **b** No. **c** Yes. **d** Yes. **e** No. **f** Yes. **g** No. **h** Yes. **i** Yes.

2 Maths Pathway, Arcs and Angles in Circles, Q3(b)–(f)

b $2y = 102°$, $y = 51°$ **c** $z = 2 \times 40° = 80°$ **d** $2x = 48°$, $x = 24°$
e $x = 2 \times 60° = 120°$, $2y = 120°$, $y = 60°$ **f** $x = 2 \times 25° = 50°$, $2z = 50°$, $z = 25°$

Reason in each: the angle at the centre is twice the angle at the circumference standing on the same arc.

3 Maths Pathway, Arcs and Angles in Circles, Q4(b)–(c)

b $y = 36°$ **c** $z = 29°$ (angles at the circumference standing on the same arc)

4 Maths Pathway, Arcs and Angles in Circles, Q6(b)–(c)

b $a = 180° - 140° = 40°$ **c** $y = 180° - 90° = 90°$ (opposite angles of a cyclic quadrilateral)

5 Maths Pathway, Arcs and Angles in Circles, Q8(b)–(d)

b $k = 31°$, because the marked angles are subtended by the same arc at points on the circle.

c $m = 180° - 108° = 72°$, because the marked angles are opposite angles in a quadrilateral whose vertices are on the circle.

d $z = 2 \times 72° = 144°$, because the marked angles are subtended by the same arc; the angle marked z is subtended at the centre, and the angle marked $72°$ is subtended at a point on the circle.

MEDIUM

6 Maths Pathway, Arcs and Angles in Circles, Q5

a $x = 2 \times 160° = 320°$ **b** $2y = 280°$, $y = 140°$ **c** $z = 360° - 60° = 300°$ **d** $a = 360° - 270° = 90°$

e $2p = 100°$, $p = 50°$; $q = 360° - 100° = 260°$ **f** $x = 360° - 120° = 240°$, $2y = 240°$, $y = 120°$

g $a = 2 \times 75° = 150°$, $b = 360° - 150° = 210°$, $2c = 210°$, $c = 105°$

h $x = 360° - 80° = 280°$, $2y = 280°$, $y = 140°$, $2z = 80°$, $z = 40°$

i $f = 2 \times 130° = 260°$, $g = 360° - 260° = 100°$, $2h = 100°$, $h = 50°$

Reasons: angles in a revolution at the centre add to $360°$, and the angle at the centre is twice the angle at the circumference standing on the same arc (a reflex angle at the centre goes with an obtuse angle at the circumference).

7 Maths Pathway, Arcs and Angles in Circles, Q2

a Since OA and OB are radii of the same circle, $OA = OB$. Therefore ΔOAB is isosceles. Since OA and OC are radii of the same circle, $OA = OC$. Therefore ΔOAC is isosceles.

b $\angle OBA = \angle OAB = x$ and $\angle OCA = \angle OAC = y$ (base angles of an isosceles triangle are equal).

c In ΔOAB, $\angle AOB + x + x = 180°$, so $\angle AOB = 180° - 2x$. In ΔOAC, $\angle AOC + y + y = 180°$, so $\angle AOC = 180° - 2y$.

d $\angle BOC + \angle AOB + \angle AOC = 360°$, so $\angle BOC = 360° - (180° - 2x) - (180° - 2y) = 2x + 2y$.

e $\angle BAC = x + y$ **f** $\angle BOC = 2x + 2y = 2 \times \angle BAC$

8 AMSI TIMES Circle Geometry, Exercise 8

a The angles $\angle DAC$ and $\angle DBC$ stand on the same arc DC, so $\angle DBC = \alpha$.

b $\angle BDC = \beta$ (angles on the same arc BC), $\angle ABD = \gamma$ (angles on the same arc AD), $\angle ADB = \delta$ (angles on the same arc AB).

c $2\alpha + 2\beta + 2\gamma + 2\delta = 360°$ (angle sum of quadrilateral $ABCD$), so $\alpha + \beta + \gamma + \delta = 180°$.

d The source says 'The rest is clear.' In full: $\angle A = \alpha + \beta$ and $\angle C = \gamma + \delta$, so $\angle A + \angle C = 180°$; then $\angle B + \angle D = 360° - 180° = 180°$.

9 AMSI TIMES Circle Geometry, Exercise 9

Join the common chord BQ, and produce ABC to X, and let $\alpha = \angle A$.
Then $\angle BQR = \alpha$ (opposite exterior angle of cyclic quadrilateral)
and $\angle RCX = \alpha$ (opposite exterior angle of cyclic quadrilateral),
so $AP \parallel CR$ (corresponding angles are equal)

The two cyclic quadrilaterals are $ABQP$ (first circle) and $BCRQ$ (second circle).

10 AMSI TIMES Circle Geometry, Exercise 11

The adjacent interior angle is supplementary to the exterior angle, and therefore equal to the opposite interior angle. Hence the quadrilateral is cyclic by the theorem.

In full: the exterior angle equals the opposite interior angle, and it is also supplementary to the adjacent interior angle, so the adjacent and opposite interior angles are supplementary. Opposite angles supplementary is the test for a cyclic quadrilateral.

11 AMSI TIMES Circle Geometry, Exercise 12

$\angle D = 180° - \alpha$ (co-interior angles, $AB \parallel DC$)
so $\angle D + \angle B = 180°$
Hence the quadrilateral $ABCD$ is cyclic (opposite angles are supplementary).

12 AMSI TIMES Circle Geometry, Exercise 6

Case 2: It will be sufficient to prove the result when O lies on AP.
Let $\angle OBP = \beta$.
Then $\angle BPO = \beta$ (base angles of isosceles $\triangle OBP$),
so $\angle AOB = 2\beta = 2 \times \angle APB$ (exterior angle of $\triangle OBP$).

Case 3: Join PO and produce PO to Q.
Let $\angle OAP = \alpha$ and $\angle OBP = \beta$.
Then $\angle APO = \alpha$ (base angle of isosceles $\triangle AOP$) and $\angle OPB = \beta$ (base angle of $\triangle OBP$).
Hence $\angle APB = \beta - \alpha$.
Also $\angle AOQ = 2\alpha$ (exterior angle of $\triangle OAP$) and $\angle BOQ = 2\beta$ (exterior angle of $\triangle OBP$).
Hence $\angle AOB = 2\beta - 2\alpha = 2(\beta - \alpha) = 2 \times \angle APB$.

HARD

13 AMSI TIMES Circle Geometry, Exercise 10

In this proof, we construct two isosceles triangles. Let $ABCD$ be a cyclic trapezium with $AB \parallel DC$. Since $ABCD$ is not a rectangle, and its angles add to $360°$, one of its angles is acute.
Let $\angle A = \alpha$ be acute, and produce BC to P.
Then $\angle DCP = \alpha$ (opposite exterior angle of cyclic quadrilateral)
so $\angle B = \alpha$ (corresponding angles, $AB \parallel DC$).
Hence $\angle B$ is also acute, so AD and BC produced meet at a point M,
and $\angle CDM = \alpha$ (corresponding angles, $AB \parallel DC$).
Hence $\triangle DMC$ and $\triangle AMB$ are both isosceles, with $DM = CM$ and $AM = BM$, so $AD = BC$.

14 AMSI TIMES Circle Geometry, Exercise 7

a The interval HC subtends right angles at P and Q, so the circle with diameter HC passes through them.

b The interval AB subtends right angles at P and Q, so the circle with diameter AB passes through them.

c They stand on the same arc HQ of the first circle.

d They stand on the same arc AQ of the second circle.

e In the triangles ABQ and ACR, $\angle ABQ = \angle RCA$ by parts **c** and **d**, and $\angle A$ is common, so $\angle ARC = \angle AQB = 90°$.

15 AMSI TIMES Circle Geometry, Exercise 13

a In Case 1, $\angle BCP = 90°$ (angle in a semicircle) and $\angle P = \alpha$ (angles on the same arc BC). Hence $\dfrac{a}{2R} = \sin\alpha$ (simple trigonometry in $\triangle BCP$), so $\dfrac{a}{\sin A} = 2R$.

b In Case 2, $\angle BCP = 90°$ (angle in a semicircle) and $\angle P = 180° - \alpha$ (opposite angles of cyclic quadrilateral $BACP$), so $\sin P = \sin(180° - \alpha) = \sin\alpha$. Hence $\dfrac{a}{2R} = \sin\alpha$ (simple trigonometry in $\triangle BCP$), so $\dfrac{a}{\sin A} = 2R$.

c In Case 3, $\alpha = 90°$ (angle in a semicircle), so $\sin\alpha = 1$. The diameter of the circumcircle is a, so $\dfrac{a}{\sin A} = \dfrac{a}{1} = 2R$.

In **a**, BP is a diameter, so $BP = 2R$ is the hypotenuse of $\triangle BCP$ and $BC = a$ is opposite $\angle P$. Part **b** uses $\sin(180° - \alpha) = \sin\alpha$ from the Trigonometry pack.

16 Sydney Boys High School, 2021, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part G (e)

i In cyclic quadrilateral $AXBO$, $\angle AOB + \angle AXB = 180°$ (opposite angles of a cyclic quadrilateral are supplementary) and $\angle AXB = 60°$ ($\triangle AXB$ is equilateral), so $\angle AOB = 180° - 60° = 120°$. In the same way, in $BYCO$, $\angle BOC = 120°$. Then $\angle AOC = 360° - 120° - 120° = 120°$ (angles at a point).

ii In quadrilateral $AOCZ$, $\angle AZC = 60°$ (angle in equilateral $\triangle AZC$), so $\angle AOC + \angle AZC = 120° + 60° = 180°$. Therefore $AOCZ$ is cyclic (opposite angles are supplementary).

iii Construct the chord OB with midpoint M. In circle $\mathcal{C}_1$, $\angle PMO = 90°$ (the line from the centre of a circle that bisects a chord is perpendicular to the chord). By the same logic in $\mathcal{C}_2$, $\angle QMO = 90°$. Since $\angle PMO + \angle QMO = 180°$, PMQ is a straight line.

Construct the chord OA with midpoint N. By the same logic, in $\mathcal{C}_1$ and in circle $AOCZ$, $\angle PNO = 90°$ and PNR is a straight line.

In quadrilateral $PMON$: $\angle NPM + \angle PMO + \angle MON + \angle ONP = 360°$ (angle sum of a quadrilateral), so $\angle NPM + 90° + 120° + 90° = 360°$ ($\angle MON = \angle AOB$, found in (i)), and $\angle RPQ = \angle NPM = 60°$.

By the same logic, using the chord OC, $\angle PQR = 60°$. Therefore $\triangle PQR$ is equilateral (all its angles are $60°$).

Year 10 Circle Geometry - Homework Answers

Lesson C2 | Angles at the centre and circumference

Past-Paper Homework - Worked Solutions

PP 1 **Sydney Boys High School, 2021, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part B (p)**

$\angle ADE$ is an exterior angle of the cyclic quadrilateral $ABCD$, so it equals the opposite interior angle: $\angle ABC = 76°$.

Or: $\angle ADC = 180° - 76° = 104°$ (straight line CDE), and $\angle ABC = 180° - 104° = 76°$ (opposite angles of a cyclic quadrilateral). Reasons are not required for this 1-mark item.

PP 2 **Sydney Boys High School, 2023, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part C Q4**

$\angle ABC = 90°$ (angle in a semicircle, AOC a diameter)
$\angle BCA = 180° - 55° - 90° = 35°$ (angle sum of $\triangle ABC$)
$\angle ADB = \angle BCA = 35°$ (angles at the circumference standing on the same arc AB)

The intersection of AC and BD is not the centre: do not assume it is.

PP 3 **Girraween High School, 2020, Year 10 Term 4 (Yearly Examination), Q25(b)**

Let the chords CA and BD meet at X.
$\angle CBD = \angle CAD = 55°$ (angles at the circumference standing on the same arc CD)
In $\triangle XBC$, $m° = \angle BXA$ is an exterior angle, so $m = 46 + 55 = 101$ (the exterior angle equals the sum of the two opposite interior angles).

Or: $\angle BDA = \angle BCA = 46°$ (same arc AB) and $m°$ is the exterior angle of $\triangle XDA$.

PP 4 **Girraween High School, 2020, Year 10 Term 4 (Yearly Examination), Q25(c)**

In $\triangle OAD$, $OA = OD$ (radii of a circle), so $\triangle OAD$ is isosceles and $\angle OAD = 40°$ (angles opposite the equal sides are equal).
$ABCD$ is a cyclic quadrilateral and $\angle DAB = \angle OAD$ (O lies on AB), so $x = 180 - 40 = 140$ (opposite angles of a cyclic quadrilateral are supplementary).

PP 5 **Sydney Boys High School, 2024, Year 10 Stage 5 Term 4 Yearly Exam, Mathematics, Part E Q2**

i In $\triangle FED$ and $\triangle FBA$:
$\angle EFD = \angle BFA = y$ (vertically opposite angles)
$\angle DEF = \angle BAF = x$ (angles on the same arc BD)
$\angle FBA = \angle DEF = x$ (alternate angles, $AB \parallel ED$)
In $\triangle ABF$: $\angle BFA + \angle BAF + \angle FBA = 180°$ (angle sum of a triangle), so $y + x + x = 180$, that is, $2x + y = 180$.
ii $x = \angle BAD$ and $\theta = \angle BCD$ are opposite angles of the cyclic quadrilateral $ABCD$, so $x = 180 - \theta$.
iii From **i**, $y = 180 - 2x = 180 - 2(180 - \theta)$, so $y = 2\theta - 180$.

Year 10 Circle Geometry - Homework Answers

Lesson C3 | Tangents and secants

Lesson Examples

C1 **Tutor demo.**

a The tangent is perpendicular to the radius, so $x + 42° = 90°$ and $x = 48°$.

b $\angle OTP = 90°$ (tangent and radius), so by Pythagoras' theorem $PT^2 = OP^2 - OT^2 = 13^2 - 5^2 = 144$, and $PT = 12$ cm.

Your turn.

a $x + 33° = 90°$ (tangent and radius), so $x = 57°$. **b** $PT^2 = 17^2 - 8^2 = 225$, so $PT = 15$ cm.

C2 **Tutor demo.**

a $\angle OBA = \angle OCA = 90°$ (tangent and radius). The angles of quadrilateral $ABOC$ add to $360°$, so $\angle COB = 360° - 64° - 90° - 90° = 116°$.

b $PA = PB$ (tangents from an external point), so $\triangle PAB$ is isosceles and $\angle PBA = \angle PAB = 66°$. Then $\angle APB = 180° - 66° - 66° = 48°$ (angle sum of a triangle).

Your turn.

a $\angle COB = 360° - 52° - 90° - 90° = 128°$ **b** $\angle PBA = 74°$, so $\angle APB = 180° - 2 \times 74° = 32°$.

C3 **Tutor demo.**

a $\angle BAU$ is an angle between the chord AB and the tangent AU, and $\angle APB$ is in the alternate segment, so $x = 58°$ (alternate segment theorem).

b y is the angle between the chord AB and the tangent AU, and the $47°$ angle at P is in the alternate segment, so $y = 47°$.

Your turn.

a $x = 64°$ **b** $y = 39°$ (alternate segment theorem)

C4 **Tutor demo.**

a $3 \times x = 6 \times 4$, so $x = 8$. **b** $10 \times y = 5 \times 8$, so $y = 4$. (products of the intercepts on intersecting chords are equal)

Your turn.

a $3 \times x = 9 \times 2$, so $x = 6$. **b** $8 \times y = 4 \times 6$, so $y = 3$.

C5 **Tutor demo.**

a $12 \times x = 16 \times 21$, so $x = \dfrac{336}{12} = 28$ (products of the intercepts on two secants from an external point are equal).

b $MA = 4 + 5 = 9$. $MT^2 = MB \times MA = 4 \times 9 = 36$, so $MT = 6$ (square of the tangent).

Your turn.

a $10 \times x = 14 \times 25$, so $x = 35$. **b** $MA = 16$, $MT^2 = 4 \times 16 = 64$, so $MT = 8$.

Homework

EASY

1 Maths Pathway, Chords, Secants and Tangents, Q7

a The angle between the tangent and the radius is $90°$. **b** The angle between the tangent and the radius is $90°$.

2 Maths Pathway, Chords, Secants and Tangents, Q8(b)

The tangent is perpendicular to the radius, therefore $y + 31° = 90°$ and $y = 59°$.

3 Maths Pathway, Intersecting Chords and Secants, Q2

a $\dfrac{HI}{KI} = \dfrac{JI}{LI}$, and $HI \times LI = JI \times KI$. **b** $f \times t = w \times d$

4 Maths Pathway, Intersecting Chords and Secants, Q3(b)–(d)

b $y \times 4 = 8 \times 3$, $y = \dfrac{24}{4}$, $y = 6$ **c** $m \times 4 = 3 \times 3$, $m = \dfrac{9}{4}$ or 2.25 **d** $k \times 3 = 6 \times 1$, $k = \dfrac{6}{3}$, $k = 2$

5 Maths Pathway, Intersecting Chords and Secants, Q5

a $\dfrac{EF}{EH} = \dfrac{EG}{EI}$, and $EF \times EI = EG \times EH$. **b** $p \times k = t \times m$

6 Maths Pathway, Intersecting Chords and Secants, Q6(b)–(d)

b $10 \times z = 12 \times 40$, $z = \dfrac{480}{10}$, $z = 48$ **c** $t \times 7 = 5.25 \times 8$, $t = \dfrac{42}{7}$, $t = 6$ **d** $n \times 36 = 24 \times 30$, $n = \dfrac{720}{36}$, $n = 20$

MEDIUM

7 Maths Pathway, Chords, Secants and Tangents, Q9(b)

Since AB is tangent to the circle, $\angle OBD = 90°$. $\angle OBC + \angle CBD = \angle OBD$, so $\angle OBC + 60° = 90°$ and $\angle OBC = 30°$.
Since $OB = OC$, $\triangle OBC$ is isosceles, so $\angle OCB = \angle OBC = 30°$. The sum of angles in a triangle is $180°$, so $\angle BOC = 180° - 30° - 30° = 120°$.
$\angle BOC$ and $\angle BOA$ form a straight angle, so $\angle BOA = 180° - 120° = 60°$.
In $\triangle OAB$, $\angle OBA = 90°$, so $\angle OAB = 180° - 90° - 60° = 30°$.

8 Maths Pathway, Intersecting Chords and Secants, Q4

a In ΔPQS and ΔPTR, we compare corresponding angles.
$\angle QRT = 180° - \angle QST$ (opposite angles in a quadrilateral whose points are on the circle), and $\angle PSQ = 180° - \angle QST$ ($\angle PSQ$ and $\angle QST$ form a straight angle).
Therefore $\angle PSQ = \angle QRT = \angle PRT$.
$\angle STR = 180° - \angle SQR$ (opposite angles in a quadrilateral whose points are on the circle), and $\angle PQS = 180° - \angle SQR$ ($\angle PQS$ and $\angle SQR$ form a straight angle).
Therefore $\angle PQS = \angle STR = \angle PTR$.
$\angle QPS$ is common to both triangles.
Therefore ΔPQS is similar to ΔPTR.

b In ΔPQS and ΔPTR, side PQ corresponds to side PT, and side PS corresponds to side PR. Therefore $\dfrac{PQ}{PT} = \dfrac{PS}{PR}$, so $PQ \times PR = PS \times PT$.

9 AMSI TIMES Circle Geometry, Exercise 14

Let tangents from an external point P touch the circle at T and U. Join the radii OT and OU and the interval OP. Since OP is common, the radii are equal, and the radii are perpendicular to the tangents,
$\triangle OPT \equiv \triangle OPU$ (RHS)
so $PT = PU$ (matching sides of congruent triangles).

With Pythagoras' theorem instead: $PT^2 = OP^2 - OT^2 = OP^2 - OU^2 = PU^2$.

10 AMSI TIMES Circle Geometry, Exercise 16

Tangents from an external point are equal, so we can label the lengths in the figure as shown. Then both sums of opposite sides of the quadrilateral are $a + b + c + d$.

In full: label the tangents from A, B, C and D as a, b, c and d (each vertex has two equal tangents). Then $AB + CD = (a + b) + (c + d)$ and $BC + DA = (b + c) + (d + a)$.

11 AMSI TIMES Circle Geometry, Exercise 17(a)

i 0 **ii** 1 **iii** 2 **iv** 3 **v** 4.

12 AMSI TIMES Circle Geometry, Exercise 18

a Use the AAS test. **b** They are matching sides of congruent triangles. **c** Use the RHS test.

d Hence $\angle LCI = \angle MCI$, so IC is also an angle bisector, thus proving the concurrence of the angle bisectors. Also the circle with centre I and radius $LI = MI = NI$ is tangent to all three sides by the radius-and-tangent theorem.

In **a**: $\angle MAI = \angle NAI = \alpha$, $\angle AMI = \angle ANI = 90°$ and AI is common. In **c**: $\angle ILC = \angle IMC = 90°$, IC is common, and $LI = MI$ by **b**.

13 AMSI TIMES Circle Geometry, Exercise 23

Proof: In the triangles AQM and PBM:
$\angle A = \angle P$ (angles on the same arc BQ)
$\angle M$ is common to both triangles
so $\triangle AQM$ is similar to $\triangle PBM$
Hence $\dfrac{AM}{PM} = \dfrac{QM}{BM}$ (matching sides of similar triangles)
So $AM \times BM = PM \times QM$.

14 AMSI TIMES Circle Geometry, Exercise 21

We have already proven that $\angle A = \angle Q$ and that $\angle P = \angle B$. The sine rule $\dfrac{x}{\sin X} = \dfrac{y}{\sin Y}$ in a triangle XYZ can also be written in the form $\dfrac{x}{y} = \dfrac{\sin X}{\sin Y}$.
Using this form of the sine rule in the two triangles,

$$\frac{AM}{PM} = \frac{\sin P}{\sin A} = \frac{\sin B}{\sin Q} = \frac{QM}{BM}.$$

HARD

15 AMSI TIMES Circle Geometry, Exercise 17(b)–(c)

b The indirect tangents AT and BU intersect at M. Then $AM = BM$ and $TM = UM$ because tangents from an external point are equal, and adding the lengths gives $AT = BU$.

c With direct common tangents, there are two cases to consider. Suppose first that the direct tangents AT and BU intersect at M when produced. Then again $AM = BM$ and $TM = UM$, and subtracting gives $AT = BU$.
Now suppose that the direct common tangents AT and BU are parallel. The radii AO and BO are perpendicular to the parallel tangents, so they lie on the one line and form a diameter, and the radii TZ and UZ also form a diameter. Hence $ABUT$ is a rectangle, so its opposite sides AT and BU are equal.

Here A, B are the points of contact on one circle (centre O) and T, U on the other (centre Z).

16 AMSI TIMES Circle Geometry, Exercise 19

In Case 2, $\angle AOB = 2\alpha$ (angles at centre and circumference on same arc AB), so $\angle OBA = 90° - \alpha$ (angle sum of isosceles $\triangle OAB$). But $\angle OBU = 90°$ (radius and tangent), so $\angle ABU = \alpha$ (adjacent angles at B).

In Case 3, reflex $\angle AOB = 2\alpha$ (angles at centre and circumference on same arc AB), so non-reflex $\angle AOB = 360° - 2\alpha$ (angles in a revolution), and $\angle OBA = \alpha - 90°$ (angle sum of isosceles $\triangle OAB$). $\angle OBU = 90°$ (radius and tangent), so $\angle ABU = \alpha$ (adjacent angles at B).

17 AMSI TIMES Circle Geometry, Exercise 20

Let $\alpha = \angle A$. Then $\angle BTU = \alpha$ (alternate segment theorem), so $\angle STP = \alpha$ (vertically opposite angles), so $\angle Q = \alpha$ (alternate segment theorem). Hence $AB \parallel PQ$ (alternate angles are equal).

18 AMSI TIMES Circle Geometry, Exercise 24

Proof: In the triangles ATM and TBM:
$\angle TAM = \angle BTM$ (alternate segment theorem)
$\angle AMT = \angle TMB$ (common)
so $\triangle ATM$ is similar to $\triangle TBM$ (AA)
$\dfrac{AM}{TM} = \dfrac{TM}{BM}$ (matching sides of similar triangles)
$AM \times BM = TM^2$

19 AMSI TIMES Circle Geometry, Exercise 25

Let the chords AB and PQ meet at M. Join SM and produce it to meet the circles at X and Y. We need to prove that the points X and Y coincide. Using the intersecting chord theorem in each circle in turn,

$$\begin{aligned} SM \times MX &= AM \times MB \text{ (from Circle 1)} \\ &= PM \times MQ \text{ (from Circle 2)} \\ &= SM \times MY \text{ (from Circle 3).} \end{aligned}$$

So $MX = MY$, and X and Y are on the same side of M, so they coincide: the line SM meets the third circle again at T, and the three chords pass through M.

20 Sydney Boys High School, 2023, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part F Q3(a)–(b)

a $\angle ADP = \angle APX$ (alternate segment theorem in $\mathcal{C}_1$: PX is the tangent at P and PA is a chord)
$\angle DPQ = \angle ADP$ (alternate angles, $AD \parallel PQ$)
$\therefore \angle APX = \angle DPQ$.

b In the same way in $\mathcal{C}_2$: $\angle BPY = \angle BCP$ (alternate segment theorem, PY the tangent at P) and $\angle BCP = \angle CPR$ (alternate angles, $BC \parallel PR$), so $\angle BPY = \angle CPR$.
$\angle APD = 90°$ and $\angle BPC = 90°$ (angles in semicircles).
$\angle XPQ = \angle YPR$ (vertically opposite angles).
$\angle XPQ = \angle APX + \angle APD + \angle DPQ = 2\angle APX + 90°$, using **a**.
$\angle YPR = \angle BPY + \angle BPC + \angle CPR = 2\angle CPR + 90°$.
$\therefore \angle APX = \angle CPR$. These are vertically opposite positions at P on the straight line XPR, so A, P and C are collinear.

This is the school's alternative solution. Part (c) of the question (show that $ABCD$ is cyclic) needs the converse of the angles in the same segment theorem, which is not taught in this pack, so it is left out.

Year 10 Circle Geometry - Homework Answers

Lesson C3 | Tangents and secants

Past-Paper Homework - Worked Solutions

PP 1 **Baulkham Hills High School, 2020, Year 10 Yearly Examination, Mathematics, Q9**

$TA = TB$ (tangents from an external point), so $\triangle TAB$ is isosceles and $\angle TAB = \dfrac{180^\circ - \theta}{2} = 90^\circ - \dfrac{\theta}{2}$.

$\angle TAB$ is the angle between the tangent TA and the chord AB, and P is in the alternate segment, so $\angle APB = \angle TAB = 90^\circ - \dfrac{\theta}{2}$ (alternate segment theorem).

The answer is **B**. A check with $\theta = 40^\circ$: then $\angle TAB = 70^\circ$, so $\angle APB = 70^\circ$, while options A, C and D give 20°, 40° and 140°.

PP 2 **Girraween High School, 2020, Year 10 Term 4 (Yearly Examination), Q16**

$BD = BC + CD = 2 + 5 = 7$. The square of the tangent equals the product of the intercepts on the secant: $AB^2 = BC \times BD = 2 \times 7 = 14$, so $AB = \sqrt{14}$.

The answer is **(B)**. Option (A) uses 2×5, the chord CD instead of the whole secant BD; (D) uses 5×7.

PP 3 **Girraween High School, 2020, Year 10 Term 4 (Yearly Examination), Q17**

The intercepts 4 and 6 lie on one chord, and 5 and x on the other. The products of the intercepts are equal: $5 \times x = 4 \times 6$, so $x = \dfrac{24}{5} = 4\frac{4}{5}$ (that is, 4.8).

The answer is **(B)**. Option (A) pairs the lengths wrongly ($6x = 4 \times 5$), and (D) uses $4x = 5 \times 6$.

PP 4 **Sydney Boys High School, 2022, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part B Q3**

$PA = PB$ (tangents from an external point), so $\angle PBA = \angle PAB = 71^\circ$ and $\angle APB = 180^\circ - 71^\circ - 71^\circ = 38^\circ$.

PP 5 **Sydney Boys High School, 2020, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics Stage 5.3, Q21**

a The 70° angle is between the tangent TB and the chord TQ, and $\alpha = \angle TPQ$ is in the alternate segment, so $\alpha = 70^\circ$ (alternate segment theorem).

b $TPQR$ is a cyclic quadrilateral, so $\beta = 180^\circ - 70^\circ = 110^\circ$ (opposite angles of a cyclic quadrilateral are supplementary).

PP 6 **Baulkham Hills High School, 2020, Year 10 Yearly Examination, Mathematics, Q20**

$x(x+3) = 4(4+6)$ (the products of the intercepts of two secants from an external point are equal)
$x^2 + 3x = 40$
$x^2 + 3x - 40 = 0$
$(x+8)(x-5) = 0$
$x = 5$ or $x = -8$, and $x > 0$ because it is a length, so $x = 5$.

PP 7 **Girraween High School, 2020, Year 10 Term 4 (Yearly Examination), Q25(a)**

$a = 2 \times 70 = 140$ (the angle at the centre is double the angle at the circumference on the same arc RQ)
$\angle ORP = \angle OQP = 90°$ (tangent perpendicular to radius)
$b = 360 - 140 - 90 - 90 = 40$ (angle sum of quadrilateral $ORPQ$)

PP 8 **Sydney Boys High School, 2024, Year 10 Stage 5 Term 4 Yearly Exam, Mathematics, Part C Q7**

Construct RP.
$SR = SP$ (tangents from an external point), so $\angle SRP = \angle SPR$ (angles opposite equal sides in a triangle)
$\angle SRP = (180° - 34°) \div 2 = 73°$ (angle sum of $\triangle SRP$)
$\angle RQP = \angle SRP = 73°$ (alternate segment theorem: angle between the tangent SR and the chord RP)
$\angle RQP = \angle QPV + \angle PVQ$ (exterior angle of $\triangle PQV$), and $\angle QPV = \angle QPT = 46°$ (given)
$\therefore \angle PVQ = 73° - 46° = 27°$

Year 10 Circle Geometry - Homework Answers

Review | Chapter review

R 1 **Sydney Boys High School, 2020, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics Stage 5.3, Q34**

$AP \times PB = CP \times PD$ (product of intersecting chords), so $6 \times 4 = 2 \times PD$ and $PD = 12$.
Join $OR \perp CD$: $CR = RD = \frac{14}{2} = 7$ (the perpendicular through the centre bisects the chord).
Join $OQ \perp AB$: $AQ = QB = \frac{10}{2} = 5$ (the perpendicular through the centre bisects the chord).
$OQPR$ is a rectangle (three right angles), so $OR = QP = 6 - 5 = 1$.
Join OD. In $\triangle ORD$, by Pythagoras, $OD^2 = OR^2 + RD^2 = 1^2 + 7^2 = 50$, so the radius is $OD = \sqrt{50} = 5\sqrt{2}$.

R 2 **Sydney Boys High School, 2021, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part F (c)**

$\angle ATB = 90°$ (angle in a semicircle, AB is a diameter)
$\angle BTP = 111° - 90° = 21°$
$\angle BAT = \angle BTP = 21°$ (angle in the alternate segment)
Or: $\angle OTP = 90°$ (radius perpendicular to tangent), so $\angle OTA = 21°$, and $\angle OAT = \angle OTA$ because $OA = OT$ (radii).

R 3 **Sydney Boys High School, 2022, Year 10 Stage 5.3 Term 4 Yearly Assessment, Mathematics, Part F Q3**

Join FE. $\angle FEA = 90°$ (angle in a semicircle). $\angle EDA = 90°$ (angle of the rectangle).
In $\triangle FEA$ and $\triangle EDA$: $\angle FEA = \angle EDA = 90°$ and $\angle FAE = \angle EAD$ (common angle), so $\triangle FEA \,|||\, \triangle EDA$ (equiangular).
$\frac{DA}{EA} = \frac{EA}{FA}$ (matching sides of similar triangles are in proportion), so $DA \times FA = 6^2 = 36$.
Let O be the centre of the semicircle and G the point where BC touches it. $OG \perp BC$ (tangent perpendicular to radius), so $CD = OG$, the radius, $= \frac{1}{2}FA$.
Area of $ABCD = CD \times AD = \frac{1}{2}FA \times AD = \frac{1}{2} \times 36 = 18$ cm^2.

R 4 **Girraween High School, 2020, Year 10 Term 4 (Yearly Examination), Q25(d)**

In $\triangle TAC$, $TA = TC$, so $\angle TAC = 25°$ (angles opposite equal sides in an isosceles triangle).
$\angle ATB = 25°$ (the angle between the tangent TA and the chord TB equals the angle $\angle TCB$ in the alternate segment)
$x = 180 - 25 - 25 = 130$ (angle sum of $\triangle ABT$)